



# RAMCO INSTITUTE OF TECHNOLOGY

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## Department of Computer Science and Engineering

Academic Year: 2023- 2024 (Odd Semester)

**Degree, Semester & Branch:** I Semester B.E. CSE

**Course Code & Title:** GE3151 Problem Solving and Python Programming

**Name of the Faculty member (s):** Mrs.M.Swarna Sudha

### Innovative Practice Description

**Unit / Topic:** Unit III / String functions and methods

**Course Outcome:** CO 3

**Topic Learning Outcome:** TLO 09

**Activity Chosen:** Learning by Teaching

**Date of Implementation:** 15.11.2023

### Justification:

String functions are fundamental concepts in programming and have wide-ranging applications. As students take on the role of teachers, they engage in discussions, answer questions, and share insights with their peers. This peer-to-peer interaction not only reinforces their own understanding but also allows for diverse perspectives and approaches to problem-solving, enriching the overall learning experience.

**Time Allotted for the Activity:** 35 minutes

### Details of the Implementation:

- Before the class, the teacher, inquired about the students' interest in teaching the string and its methods few students expressed their interest.
- Based on their interest Athish M Arjun. S and Deepalakshmi M were given topics on both string and its methods .
- The adequately time was given to the students to prepare for their teaching role so the topic was given a day before the class.
- Students presented their allocated topics to their classmates
- To keep the class interested, students utilized a variety of instructional strategies. Athish employed real-world examples of strings in use, such as applications, programming scenarios, or data manipulations, to demonstrate the practical relevance of the concept. Arjun used visual aids like flashcards with visual elements to aid in quick recall of key string manipulation methods, including substring operations and concatenation. Deepalakshmi used animated videos that visually represent how string methods function, offering a dynamic and interactive way for students to grasp abstract concepts related to manipulating text data..
- To promote involvement, students were allowed to ask questions to their peers during

the presentation. Because of this, students were able to have conversations and share their perspectives about topic which very important in programming

#### CO – PO / PSO mapping:

| CO   | PO1 | PO2 | PO4 | PO9 | PO10 | PSO1 |
|------|-----|-----|-----|-----|------|------|
| CO 3 | 2   | 2   | 1   | 1   | 1    | 1    |

(1 – Low 2 – Moderate 3 High)

#### PO / PSO mapped:

| Innovative practice                  | PO1                                                                                               | PO2                                                                                  | PO4                                                                                                                             | PO9                                                                                | PO10                                                         | PSO1                                                                                                            |
|--------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
|                                      | 2                                                                                                 | 2                                                                                    | 1                                                                                                                               | 1                                                                                  | 1                                                            | 1                                                                                                               |
| <b>Justification for correlation</b> | Apply mathematical concepts to analyze and design solutions related to Strings and their methods. | Develop the ability to identify engineering problems related to string manipulation. | Analyze the problem to find solutions for string-related issues such as substring operations, concatenation, and other methods. | Function effectively as an individual, and as a member or leader in diverse teams, | To communicate effectively on complex engineering activities | To excel in software development, effectively to solve complex computational tasks related to strings in Python |

#### Images / Screenshot of the practice:



**ATHISH M**



**DEEPALAKSHMI M**



**ARJUN S**

**Figure 1: Teaching the Concepts of String and its Methods**

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- All the students were eagerly listened the class. Students felt they are able to understand the topic.
    - The students were able to remember key details and concepts more easily
    - The students express that role of a teacher boost their confidence in knowledge of IP addresses
    - Students highlighted that this practice helped them improve their communication skills, such as clarity, organization, and the ability to present complex ideas in a simplified manner.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Students actively engaging in teaching their peers gain a more profound understanding of the IP address
    - Actively participating in the teaching process helps students remember key details and concepts better compared to passive learning methods
    - This activity helped the student to improve communication skills.
    - Students spent their time in self-learning.
    - This activity encouraged the students to share their knowledge with others.

- ❖ **Challenges faced in implementation:**

- Effectively motivate students who are not participating in the activity due to shyness, fear of public speaking, or lack of confidence. Motivate the students to overcome their shyness and engage in active learning.
    - Implementing Learning by Teaching required additional time compared to traditional teaching methods. Planning and executing interactive teaching sessions was challenging.

**References:**

1. <https://effectiviology.com/protege-effect-learn-by-teaching>
2. [https://www.researchgate.net/publication/351905566\\_Impact\\_of\\_Seminars\\_on\\_Student\\_Soft\\_Skills\\_Development](https://www.researchgate.net/publication/351905566_Impact_of_Seminars_on_Student_Soft_Skills_Development)
3. [https://www.ritrjpm.ac.in/images/computer-science/2022-023/4Collaborative\\_Coding\\_DBMS\\_KVS.pdf](https://www.ritrjpm.ac.in/images/computer-science/2022-023/4Collaborative_Coding_DBMS_KVS.pdf)